

Lemma.

Let K/F be a field extension, and let $\alpha \in K$ be algebraic over F . Then, for any $\sigma \in \text{Aut}(K/F)$, $\sigma\alpha$ is a root of the minimal polynomial for α over F .

Proof. Let $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \in F[x]$ be the minimal polynomial for α . Then, $f(\alpha) = 0$, so $\sigma(f(\alpha)) = 0$. (The field homomorphism preserves zero).

Now, $0 = \sigma(f(\alpha)) = \sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_1\alpha + a_0)$

$$\begin{aligned} \sigma \text{ fixes } F \quad \hookrightarrow &= (\sigma\alpha)^n + \sigma(a_{n-1}) \cdot (\sigma\alpha)^{n-1} + \dots + \sigma(a_1) \cdot (\sigma\alpha) + \sigma(a_0) \\ &= (\sigma\alpha)^n + a_{n-1}(\sigma\alpha)^{n-1} + \dots + a_1(\sigma\alpha) + a_0 \\ &= f(\sigma\alpha). \end{aligned}$$

Prop. Let $H \leq \text{Aut}(K)$ be a subgroup of the group of all automorphisms of K .

Then, the collection F of elements of K fixed by all the elements of H is a subfield.

Proof. First, given $\sigma \in \text{Aut}(K)$, the set $F_\sigma = \{a \in K \mid \sigma(a) = a\}$ is subfield, because:

Let $a, b \in F_\sigma$ be given. Then, $\sigma(a-b) = \sigma(a) - \sigma(b) = a-b$, and if $b \neq 0$, then $\sigma(a \cdot b^{-1}) = \sigma(a) \cdot \sigma(b)^{-1} = a \cdot b^{-1}$, so the claim is proved.

And now, since the arbitrary intersection of fields is field, so proved. ~~QED~~
Such a subfield is called the fixed field of $S \subseteq \text{Aut}(K)$.

Prop. Let K be a field. Then,

If $F_1 \leq F_2 \leq K$, then $\text{Aut}(K/F_2) \leq \text{Aut}(K/F_1)$.

If $H_1 \leq H_2 \leq \text{Aut}(K)$, then $F_{H_2} \leq F_{H_1}$.

In other say, The inclusion relation of automorphisms and fixed field is reverse.

Lemma.

Def. A finite extension K/F is called a Galois Extension if:

$$|Aut(K/F)| = [K:F].$$

In such a case, denote $Gal(K/F) = Aut(K/F)$.

Def. A character χ of a group G with values in a field L is a group homomorphism

$$\chi: G \rightarrow L^\times$$

The characters $\chi_1, \dots, \chi_n$ of G are said to be linearly independent over L if:

$$\alpha_1 \chi_1 + \dots + \alpha_n \chi_n \equiv 0 \text{ for some } \alpha_i \in L, \text{ then } \alpha_1 = \dots = \alpha_n = 0.$$

Thm. [Dedekind-Independence of Characters Theorem].

If $\chi_1, \dots, \chi_n: G \rightarrow L^\times$ are distinct characters of G with values in L , then they are linearly independent over L .

Proof. Suppose that the characters are linearly dependent.

That is, there exists non-trivial representation: for some $\alpha_1, \dots, \alpha_n \in L$,

$$\alpha_1 \chi_1 + \dots + \alpha_n \chi_n \equiv 0 \text{ where } \alpha_1, \dots, \alpha_n \text{ are not all zero.}$$

So, the set $\{m \in \mathbb{N} \mid \text{for some } b_1, \dots, b_m \in L \setminus \{0\}, \text{ and } \sigma \in S_n, b_1 \chi_{\sigma(1)} + \dots + b_m \chi_{\sigma(m)} \equiv 0\}$ is non-empty, thus it has a minimal element by the well-ordering principle.

$m \in \mathbb{N}$ with $b_1 \chi_1 + \dots + b_m \chi_m \equiv 0$, the indices are renumbered, $b_i \in L \setminus \{0\}$.

Since the characters are distinct, $\chi_1(g_0) \neq \chi_m(g_0)$, for some $g_0 \in G$.

Given $g \in G$, observe that

$$0 = b_1 \chi_1(g_0 g) + \dots + b_m \chi_m(g_0 g) = b_1 \chi_1(g_0) \chi_1(g) + \dots + b_m \chi_m(g_0) \chi_m(g)$$

Meanwhile,

$$0 = \chi_m(g_0) \cdot 0 = \chi_m(g_0) \cdot (b_1 \chi_1(g) + \dots + b_m \chi_m(g)) = b_1 \chi_m(g_0) \chi_1(g) + \dots + b_m \chi_m(g_0) \chi_m(g),$$

Subtracting gives $[\chi_m(g_0) - \chi_1(g_0)] \cdot b_1 \chi_1(g) + \dots + [\chi_m(g_0) - \chi_m(g_0)] \cdot b_m \chi_m(g) = 0$, because the last term is terminated. But, since $[\chi_m(g_0) - \chi_1(g_0)] \neq 0$, it is another representation, but that is contradiction to the minimality of m .

Corollary. Distinct Field Automorphisms are linearly independent.

[Thm.

Let $G = \{\sigma_1 (= \text{id}), \sigma_2, \dots, \sigma_n\}$ be a subgroup of $\text{Aut}(K)$, and let $F = K^G$.

Then, $[K:F] = |G| = n$.